

Hospital Length of Stay (LOS) Optimization - Portfolio Case Study Brief

1. Job posting - Hospital LOS Case

Study Title

Hospital Length of Stay (LOS) Optimization - Notebook, Dashboard C GitHub Portfolio Project

Overview

I'm a pharma/clinical research professional building a healthcare analytics portfolio. For this project, I want a **case-study-style analysis** of hospital length of stay (LOS), not just a notebook

dump. The end product should show: *Which patients are likely to have long hospital stays, why, and what operations should do about it.*

Problem to solve

Hospitals face bed capacity pressure because a small group of long-stay patients consumes a disproportionate share of bed days. The goal of this project is to identify which patients are most likely to have long stays, understand the drivers of extended LOS, and recommend where care management and operations should focus discharge planning efforts.

What you will deliver

- **One Jupyter notebook** that includes:
 - Data cleaning and EDA of LOS
 - Short vs long-stay classification (e.g., LOS bucketed into short/long or short/medium/long)
 - Model evaluation and feature importance
- **2-3 SQL (or SQL-style) queries** to:
 - Build the inpatient cohort
 - Calculate basic LOS statistics (distribution, LOS by diagnosis/service line/admission type)
- **One focused dashboard page** (Streamlit, Power BI, or Tableau) showing:
 - LOS distribution and long-stay share
 - Long-stay risk by key dimensions (diagnosis group, service line, admission type)
- **GitHub repo** with:
 - README.md (problem, data, methods, key findings, recommendations)
 - Notebook, SQL files, dashboard code/BI file, and requirements.txt/environment file
- **1-2 page case summary** (Word/Google Doc or slide outline) with:
 - Problem and approach (high-level)
 - 3-5 key findings

- 3-5 “So what?” recommendations

Key findings I expect (you fill in with real numbers) Please structure the main results like this:

1. X% of admissions account for Y% of total bed days.
2. Top 3 diagnosis groups/service lines with the highest long-stay risk.
3. Relative long-stay risk for emergency + ICU admissions vs elective admissions.
4. Top 3-5 predictors from feature importance (e.g., age, comorbidity count, admission type).
5. Model performance (accuracy/F1/etc.) in one short paragraph.

Recommendations (“So what?”) I want you to propose

Please propose 3-5 specific, actionable recommendations, for example:

- Prioritize early discharge planning for the highest-risk diagnosis groups/service lines.
- Flag predicted long-stay admissions at intake for care management review.
- Use predicted long-stay risk to support bed-capacity planning by service line.

I will do the final clinical/wording polish; you just need to get the structure, logic, and content in place.

Tech stack

- SQL (or SQL-like data manipulation)
- Python (pandas, scikit-learn, plotting, plus a tree-based model like XGBoost/Random Forest)
- One of: Streamlit, Power BI, or Tableau

Budget & timeline

- Fixed price: **200-300 USD**
- Timeline: **7-14 days**

I prefer to structure this as 3 milestones:

1. Data + EDA plan
2. Modeling + draft insights
3. Dashboard + GitHub + summary

If this project goes well, I have **two more similar healthcare case studies** (appointment no-shows and cost/claims variation) and would like to continue with the same freelancer.

What to include in your proposal

- 1-2 sentences on your **healthcare or tabular ML** experience

- 1 GitHub repo with a **notebook** + **README**
 - 1 example **dashboard** (screenshot or link)
 - Your preferred tools (Python/R, Streamlit/Power BI/Tableau)
 - Confirmation that you can deliver notebook + SQL logic + dashboard + GitHub + summary within 7-14 days
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2. Master brief - LOS

version Portfolio vision

This LOS project is one of three healthcare analytics case studies I'm building (LOS, appointment no-shows, cost variation). Each project must show **SQL + Python + BI + a clear recommendation story** that I can walk through in interviews.

Common structure for this LOS project

- **Notebook**
 - Introduction: business context and problem
 - Data: description and cleaning steps
 - EDA: LOS distribution, LOS by diagnosis/service line/admission type
 - Modeling: short vs long-stay classification, metrics, feature importance
 - Interpretation: what drives long stays, limitations
- **SQL (or SQL-style)**
 - Cohort query for inpatient admissions
 - LOS distribution and summary stats
 - Optional: long-stay outlier identification
- **Dashboard**
 - 1 page, focused on:
 - LOS distribution
 - Long-stay share by key dimensions
 - Simple view of predicted long-stay risk
- **GitHub**

- Clear structure:
 - /sql/
 - /notebooks/
 - /dashboard/ (or BI file)
 - README.md
 - requirements.txt or environment file
- README written for a **non-technical stakeholder** (clinical/operations leader)
- **Case summary**
 - 1-2 pages or slide outline covering:
 - Context and problem
 - Approach (high-level)
 - 3-5 key findings in plain English
 - 3-5 clear, actionable recommendations

Fairness and Ethics Considerations

Race and marital status were included in the admission-time feature set for exploratory modeling and therefore appear in the logistic regression coefficient table, but in this project they are treated as sensitive, non-causal proxy variables rather than clinical explanations of long-stay risk. Logistic regression is retained mainly as a transparency baseline while the final operational recommendation uses Random Forest; in a real deployment, any use of sensitive demographic fields would require explicit governance, subgroup performance and error-rate testing, proxy/bias review, documentation of intended use, ongoing drift and fairness monitoring, and a human-in-the-loop workflow so the model can only trigger supportive care-management review, not restrictive or punitive decisions. If those variables did not improve performance without worsening disparities, they should be removed from the production model.

External Benchmark Context

The final Random Forest achieved a ROC-AUC of 0.73 on this demo cohort, which is slightly below but still broadly in line with published prolonged-LOS studies on much larger cohorts using different LOS definitions, such as AUC 0.754 in a 12,858-patient emergency-admission study and AUC 0.810 in a 73,182-hospitalization French administrative cohort, so this result should be interpreted as a promising but preliminary benchmark rather than a like-for-like comparison.

Benchmark sources:

[French 73,182-hospitalization study](#)[12,858-patient ED study](#)

My role vs your role

- **You:** Build the analysis, models, dashboard, GitHub repo, and first-draft summary.
 - **Me:** Final clinical/operational sense-check and wording polish, then I will present it as part of my portfolio.
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